

Vijay Kumar Singh

PhD Candidate

Computer Science & Engineering

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Research Interests

Wireless and Ubiquitous sensing for human-centric applications; multimodal sensing systems (WiFi CSI, video, HCI); Robust Human Activity Recognition; Integrated Sensing and Communication for future WiFi networks.

Education

2021–2026 Ph.D. in Computer Science and Engineering, IIIT-Delhi, New Delhi, India

CGPA: 8.68

2012–2014 M.Tech in Computer Science and Engineering (Mobile Computing), NIT Hamirpur, India

CGPA: 8.37 (*Gold Medal*)

2008–2012 B.Tech in Computer Science and Engineering, PSIT Kanpur, India

Percentage: 75.74%

Awards and Achievements

- Awarded travel support from ACM India to present the research work at Academic Research and Careers for Students (ARCS) 2026, held at IIT Hyderabad
- Awarded travel support from Google India to attend Google India Networking Research Summit at Bengaluru, India, 2026
- Awarded UbiComp student travel grant to attend UbiComp 2025, physically held in Espoo, Finland
- Awarded Overseas Research Fellowship (ORF) worth 15720 USD from IIIT Delhi for visiting scholar at University of Maryland, College Park, USA, 2025
- Awarded registration grant worth 700 USD at MobiSys 2025, held at Anaheim, California, USA
- Awarded registration grant worth 6000 INR at COMPASS 2024
- Awarded best poster runner-up award (received 5000 INR) at COMSNETS 2023
- Awarded student travel grant worth 14000 INR and accommodation to attend COMSNETS 2023.
- Awarded student travel grant worth 14000 INR and accommodation to attend COMSNETS 2022.
- Qualified for national level UGC-NET JRF/SRF PhD Research fellowship 2021-2026
- Awarded Gold Medal, M.Tech 2014
- Awarded MHRD Scholarship to pursue M.tech, National Institute of Technology, Hamirpur 2012-2014

Selected Publications and Patents

- WiLight: A Lightweight and Adaptive WiFi Sensing Framework for Real-Time HAR, IEEE Transactions of Mobile Computing (IEEE TMC, 2026 - Under Review)
- Harnessing WiFi Sensing for Video-less Assessment of Participants' Engagement during Online Synchronous Meetings, ACM Transactions on Sensor Networks (ACM TOSN, 2026)
- Slim-Sense: A Resource Efficient WiFi Sensing Framework towards Integrated Sensing and Communication, Proceedings of the ACM Transaction on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT Journal, 2026), Presented in UbiComp 2025, held in Espoo, Finland, DOI: 10.1145/3712271
- Engagement in Online Participation by Harmonizing WiFi and Audio, ACM International Conference on Multimodal Interaction (ICMI 2023), held in Paris, France, DOI: 10.1145/3577190.3614108.
- Demonstration of WIFITUNED: Monitoring Engagement in Online Participation by Harmonizing WiFi and Audio, Demo Paper in Proceedings of International Conference on COMMunication Systems & NETworks (COMSNETS 2024) held in Bangalore, India, DOI: 10.1109/COMSNETS56262.2023.10041341
- Monitoring engagement in online classes through WiFi CSI, Proceedings of International Conference on COMMunication Systems & NETworks (COMSNETS 2024) held in Bangalore, India (COMSNETS 2023), DOI: 10.1109/COMSNETS56262.2023.10041341

- CoDel-ACT: Realizing CoDel AQM for Programmable Switch ASICs, IFIP/IEEE Networking 2024, held in Thessaloniki, Greece, DOI: 10.23919/IFIPNetworking62109.2024.10619729
- An optimized multi-channel MAC scheme with dynamic control channel interval in dense VANET, ICACC 2019, held in Chennai, India, DOI: 10.1016/j.procs.2018.10.444
- An optimized multichannel MAC scheme with dynamic control channel interval in dense VANET, International Journal of Information Technology (IJIT 2019), DOI: <https://doi.org/10.1007/s41870-019-00292-z>

Patents

- System and Method for Integrated Sensing and Communication in a Wireless Fidelity Network, Indian Patent Application Number: 202511056716.
- System and Method for Multimodal Proctoring using Behavioral and Environmental Sensing, Indian Patent Application Number: 202641080080.

Academic Research Experience

July 2021 to Present **Ph.D. Research, IIIT Delhi, New Delhi**

Ph.D. Research Focus: Towards Enabling Ubiquitous, Robust, Lightweight, and Resource-Efficient WiFi Sensing.

My PhD thesis focuses on enabling ubiquitous, robust, lightweight, and resource-efficient WiFi sensing that is practically real-time deployable on existing WiFi infrastructure while coexisting with data communication in next-generation wireless networks. The thesis addresses key challenges in WiFi sensing, including robust sensing under real-world environmental variations, continuous real-time inference, efficient deployment across resource-constrained and heterogeneous edge devices, generalization to unseen users and environments, and efficient spectrum utilization for the coexistence of sensing and communication. The thesis begins with WiFiTuned, a multimodal engagement assessment system that integrates WiFi CSI-based motion sensing with audio cues to infer user engagement in online meetings. This work demonstrates the practical feasibility of continuous, real-world WiFi sensing by correlating fine-grained non-verbal behavioral cues with meeting intent. Next, the thesis focuses on large-scale, practical deployment for real-time inference of user activities across diverse sensing locations and computing platforms. It introduces an efficient signal preprocessing to mitigate environmental noise and extract Doppler traces as robust activity representation. Next, it introduces a lightweight, Doppler-aware unified learning architecture that uses reparameterizable depthwise separable convolutions and linear attention to efficiently capture multi-scale motion patterns. Device-aware structured pruning reduces model complexity according to the computational constraints of different edge devices, while a few-shot adaptation mechanism adapts the pruned model to unseen environments and users with minimal labeled data, enabling real-time activity recognition across heterogeneous computing platforms. Building on the resource demands of continuous WiFi sensing applications, the thesis next addresses efficient spectrum and antenna resource utilization through Slim-Sense, an ISAC framework that enables accurate WiFi sensing without utilizing the full available bandwidth and antenna resources. Slim-Sense enables adaptive, environment- and application-specific selection of sensing resources, reducing sensing resource utilization while preserving communication capacity and maintaining sensing accuracy.

April 2025 to September 2025 **Visiting Scholar, University of Maryland, College Park, USA.**

Research Focus: Wireless Location-Aware Gesture Tracking and Interaction with Smart Environment.

The research focuses on per-person gesture recognition and spatial localization in multiperson environments using wireless radio signals. Location cues are first extracted from range-dependent reflections and then used to obtain location-specific micro-Doppler and phase features. These features are jointly modeled using a Doppler-aware neural beamforming architecture for accurate gesture recognition and localization. A diverse dataset with up to 8 activities and 8 persons is collected and simulated across different environments to evaluate robustness and generalizability. Overall, the proposed framework enables simultaneous gesture recognition and localization without wearable devices or cameras.

July 2012 to August 2014 **M.Tech. Research, NIT Hamirpur, Himachal Pradesh.**

M.Tech. Research Focus: Multichannel MAC Scheme to Deliver Real-Time Safety Packets in VANETs.

My M.Tech thesis focuses on the design of a multichannel Medium Access Control (MAC) protocol for vehicular wireless access environments (WAVE), targeting reliable and low-latency communication in Vehicular Ad Hoc Networks (VANETs). The work is grounded in the IEEE 1609.4 standard and addresses the challenge of efficient channel coordination under dynamic vehicular traffic conditions. The thesis proposes a dynamic Control Channel Interval (DCI) MAC scheme that adapts channel access to prioritize the timely delivery of real-time safety messages while preserving bandwidth for non-safety services. By dynamically adjusting control and service channel usage, the proposed scheme enables VANETs to handle varying

traffic loads, ensuring reliable and efficient transmission of safety-critical packets for applications such as driver assistance and collision avoidance.

Research Impact

- WiFiTuned: Harnessing WiFi Sensing for Video-less Assessment of Participants' Engagement during Online Synchronous Meetings — Collaboration with IIT Kharagpur**
Developed a video-less engagement assessment of users in the online synchronous meeting with 86% accuracy, fusing WiFi signals and meeting audio — directly applicable to EdTech platforms, virtual meeting applications, and online learning analytics (COMSNETS 2023, ACM ICMI 2023, COMSNETS 2024, ACM TOSN 2026).
- Enabling Coexistence of Sensing and Communication for Integrated Sensing and Communication (ISAC)**
Developed Slim-Sense, a resource-efficient WiFi sensing framework that intelligently selects the most informative bandwidth and antenna resources, achieving 90% resource savings with only a 4% reduction in sensing accuracy — enabling spectrum-efficient WiFi sensing for seamless coexistence of sensing and communication in ISAC networks (ACM IMWUT/UbiComp 2025, Patent 2026).
- WiLight: A Lightweight and Adaptive WiFi Sensing Framework for Real-Time HAR**
Developed a lightweight and adaptive WiFi sensing framework for real-time human activity recognition, achieving up to 95% accuracy with real-time inference through robust CSI processing, Doppler-aware learning, device-aware pruning, and few-shot adaptation — relevant to smart-home automation, elderly care, device-free human-computer interaction, and safety-critical monitoring (under review).
- Wireless Location-Aware Gesture Tracking and Interaction with Smart Environment — Collaboration with UMD, College Park, USA**
Developed a wireless sensing framework for simultaneous per-person gesture recognition and spatial localization in multiperson environments, jointly exploiting range, micro-Doppler, and phase variations through Doppler-aware neural beamforming, achieving robust recognition across multiple users and environments — relevant to smart-home automation, multi-user human-computer interaction, retail analytics, and intelligent building systems (ongoing).
- Multimodal Proctoring with WiFi CSI for Out-of-View Behavioral Sensing — Collaboration with IIT Madras**
Designing a multimodal sensing system for online exams proctoring, fusing WiFi CSI, video, and device interaction logs — directly applicable to EdTech platforms, remote proctoring, and institutional assessment integrity (ongoing, Patent 2026)

Technical Expertise

- Core expertise applicable to smart ecosystems, Next-generation ISAC networks, EdTech platforms, and Smart wireless sensing
- AI/ML & Data Science:** PyTorch, TensorFlow, scikit-learn, CNNs, TCN, Vision Transformer, multimodal fusion (video + RF + HCI), SciPy, NumPy, Pandas, MediaPipe
 - Programming:** C/C++, Java, P4, Python
 - Wireless Systems:** WiFi PHY, CFR/CSI processing, OFDM, OFDMA, MIMO, Beamforming
 - Signal Processing:** Time-frequency analysis, FFT, STFT, 3D FFT, Doppler estimation, Signal Denoising, 2D-CFAR
 - Systems & Tools:** Embedded Systems, ESP32, Jetson, Raspberry Pi, Linux, Git, LaTeX, PicoScenes, ESP32 CSI Toolkit, Nexmon

Professional Experience

- Teaching Assistant | *Indraprastha Institute of Information Technology, Delhi* (August'25-December'25)
- Teaching Assistant | *Indraprastha Institute of Information Technology, Delhi* (August'24-December'24)
- Teaching Assistant | *Indraprastha Institute of Information Technology, Delhi* (January'22-May'22)
- Teaching Assistant | *Indraprastha Institute of Information Technology, Delhi* (August'21-December'21)
- Assistant Professor | *KIET Group of Institutions, Ghaziabad* (August'20-July'21)
- Assistant Professor | *DIT University, Dehradun* (August'19-July'20)
- Assistant Professor | *Chandigarh University, Mohali* (July'16-July'19)

- Assistant Professor | *SIRDA Group of Institutions, Mandi*

(Jan'15-July'16)

Internship and Training

- **Visiting Research Scholar - 6 Months** | *University of Maryland, College Park, USA* (April'25)
 - Worked on a project on multi-person gesture recognition utilizing wireless signals in the iCOSMOS Lab, Department of Computer Science and Engineering, under the guidance of Dr. Nirupam Roy
- **Visiting Research Scholar - 6 Months** | *IIT, Madras, India* (January'26)
 - Worked on a project on multimodal proctoring with WiFi CSI for out-of-view behavioral sensing in the WinSys Lab, Department of Computer Science and Engineering, under the guidance of Dr. Mukulika Maity and Dr. Ayon Chakraborty
- **Industrial Training - 5 Weeks** | *EPAM India, Hyderabad* (May'19)
 - Five Weeks Hands-on training on application development using industry Best Practices and Principles
- **Wipro Certified Trainer Program - 2 months** | *WIPRO, Noida* (June'19)
 - WIPRO Certified Trainer Certification for WIPRO Talent Next Program
- **Industrial Training - 1 Week** | *Google, Gurugram* (August'17)
 - Hands-on android development

Talks/Tutorials/Exhibitions

- Conference paper presentations: UbiComp 2025, ICMI 2023, ICACC 2018
- Poster presentation and demo exhibition: ARCS 2026, COMSNETS 2023, COMSNETS 2024, COMPASS 2024
- Project Exhibition: India Mobile Congress 2023, 2024
- Hands-on Session: Introduction to WiFi Sensing at ACM Winter School 2023, IIIT Delhi

Research Service

- TPC Member for the IEEE ANTS 2026 Conference, to be held at IIT Roorkee.
- Reviewed research papers at NCC 2026, IEEE Transactions on Mobile Computing (TMC) 2026.
- Reviewed research paper at INFOCOM 2025.
- Reviewed research paper at the Pervasive and Mobile Computing International Journal 2024.
- Student Volunteer in organizing committee at MobiSys 2025 conference.
- Student Volunteer in organizing committee at COMPASS 2024 conference.
- Student Volunteer in organizing committee in LastMiles workshop at COMSNETS 2022 conference.

References

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